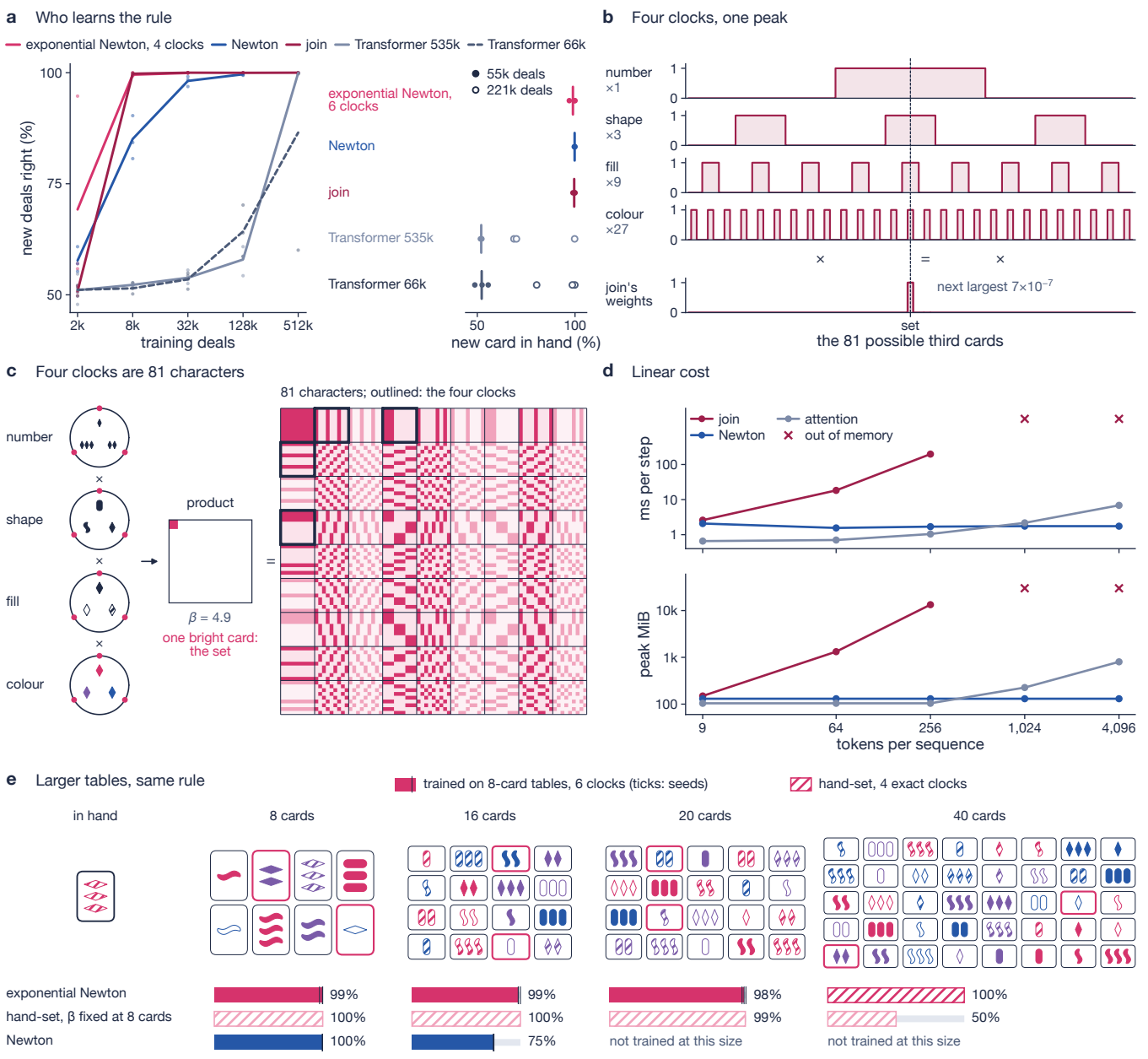

a
Who learns the rule
exponential Newton, 4 clocks
Newton
join
Transformer 535k
Transformer 66k
new deals right (%)
100
75
50
2k
8k
32k
128k
512k
training deals
55k deals
221k deals
exponential Newton, 6 clocks
Newton
join
Transformer 535k
Transformer 66k
50
100
new card in hand (%)
b
Four clocks, one peak
number ×1
shape ×3
fill ×9
colour ×27
join's weights
next largest 7×10⁻⁷
set
the 81 possible third cards
c
Four clocks are 81 characters
number
shape
fill
colour
product
β = 4.9
one bright card: the set
81 characters; outlined: the four clocks
d
Linear cost
join
Newton
attention
out of memory
ms per step
peak MiB
9
64
256
1,024
4,096
tokens per sequence
e
Larger tables, same rule
trained on 8-card tables, 6 clocks (ticks: seeds)
hand-set, 4 exact clocks
in hand
8 cards
16 cards
20 cards
40 cards
exponential Newton
hand-set, β fixed at 8 cards
Newton
99%
100%
100%
99%
100%
75%
98%
99%
not trained at this size
100%
50%
not trained at this size